

Corporation, Electronics Sector Skills Council of India, AICTE, and leading universities.

Without closing the skills gap, India risks importing both technology and talent. With the right academic-industry-policy bridge, it can instead export capability and leadership in this high-growth domain.

## Global vs local: Where India stands today

Globally, the automotive display market is dominated by Korea, Japan, Taiwan, and increasingly China, where decades of investment in display fabrication, precision materials, and high-reliability electronics have created vertically integrated ecosystems.

India, by contrast, remains largely downstream in the value chain, acting primarily as an importer, integrator, and end-user. The absence of active-matrix display fabs, specialised materials supply such as polarisers, colour filters, and driver ICs, and certified display design houses limits participation in high-value segments. Even EMS firms assembling display modules often rely on pre-certified kits, constraining innovation and local intellectual property development.

However, India holds distinct advantages that can be harnessed:

- A large and fast-growing domestic auto market with an increasing appetite for smart cockpits
- A vibrant design and software talent pool, capable of building innovative HMI, UI, and embedded display solutions
- Competitive EMS capabilities that can evolve with the right incentives and certifications
- A supportive policy framework under the ISM, Make-in-India, and the emerging display PLI

To compete globally, India must localise more of the value chain—not only assembly, but also materials research, display controller design, and qualification infrastructure. Public-private partnerships, industry consortia, and

SID-facilitated global collaborations can play a catalytic role.

## Vision 2030: What needs to happen now

The next five years are critical if India is to emerge as a global hub for automotive displays. The transition must be from passive consumption to active contribution and co-creation, anchored in a growing automotive market, EMS depth, and technical talent. Here is what India must do, starting now:

**1. Invest in display module clusters near auto hubs.** Establish display integration parks in Pune, Manesar, Sanand, and Chennai, offering cleanroom facilities, bonding lines, and testing infrastructure for automotive displays. These should support startups, EMS firms, and OEMs alike.

**2. Localisation roadmap for key components and subsystems.** Support domestic development of cover glass, optical films, driver IC packaging, and automotive-grade backlights. Encourage component partnerships through joint ventures and FDI from display leaders in Korea, Japan, and Taiwan.

**3. Create automotive display standards and certification ecosystems.** Build government-supported certification labs to test displays for vibration, temperature cycling, humidity, sunlight readability, EMI, and functional safety, aligned with global automotive standards.

**4. Launch a national display design and skills mission.** Under NSDC and AICTE, roll out display-centric skilling programmes, covering embedded HMI systems, UI/UX, and optics. Include faculty fellowships and SID-curated content for engineering colleges.

**5. Seed innovation and IP in HMI and UX.** Fund startups and research labs working on next-gen cockpit experiences, gesture interfaces, voice-first systems, driver attention monitoring, and AR-based HUDs. Encourage public-private

innovation consortia involving SID, academia, and industry.

**6. Establish a display sub-mission within Make-in-India.** Introduce a focused sub-policy aligned to automotive and ISM programmes, addressing display modules, components, tooling, and validation. Offer long-term clarity and de-risking incentives for strategic investors.

If these strategic moves are implemented, by 2030 India could not only meet its domestic automotive display demand indigenously but also emerge as a Tier-1 export partner for global auto OEMs. The roadmap is clear; what is needed is coordinated execution, global engagement, and national resolve.

## India's moment is now

Automotive displays are no longer futuristic embellishments. They are the core interface of modern mobility, shaping how drivers interact with vehicles, data, and the environment. As vehicles become electric, connected, and increasingly software-defined, the cockpit is emerging as the command centre, with displays at its heart.

India's strong automotive base, capable EMS sector, and deep design talent position it uniquely to shape this future. The strong response to the SID India Automotive Display and Electronics Workshop at the India Habitat Centre in New Delhi highlighted readiness across industry, academia, and policy. Yet opportunity windows close quickly. Without timely intervention, India risks long-term dependence on imports while others capture value, jobs, and intellectual property.

The way forward lies in aligning automotive demand, EMS capabilities, and support from the Make-in-India policy to build a globally competitive, locally anchored automotive display ecosystem. The road ahead is complex, but the direction is clear. If India acts decisively and collaboratively now, it can lead—not follow—in the global race for intelligent, display-driven mobility. **EFY**